

Outset · AI in Healthcare · Day 3 · Group 4

Good for whom?

Skin-cancer AI, access, and who is missing from the training data

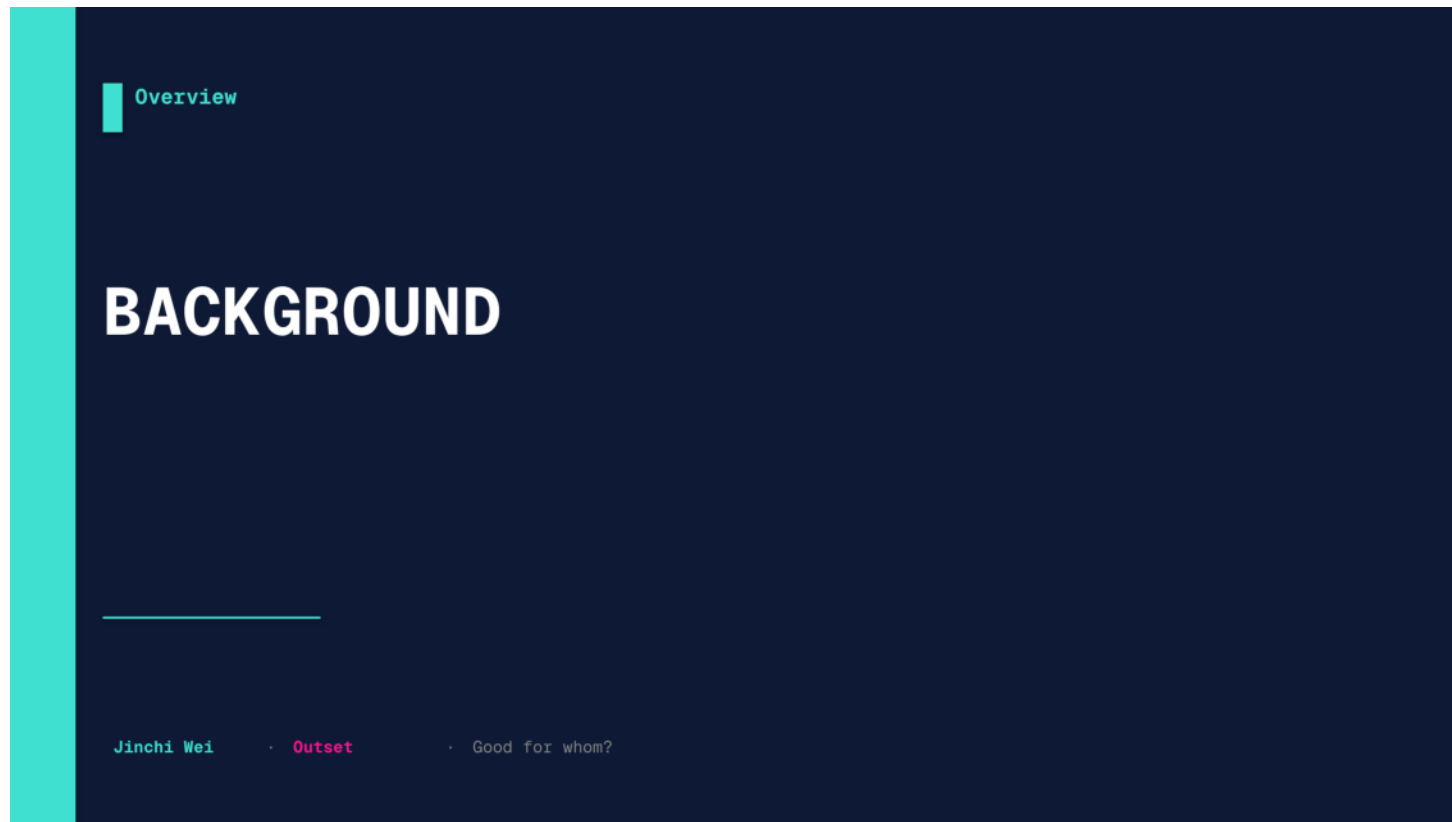
Jinchi Wei

Outset

2026-07-08



Slide 1 -



Slide 2 – Background

This section sets up the tension before any of our own results: skin-cancer AI is sold as a way to widen access, yet it learns from images that mostly show light skin. Prime students to ask 'good for whom?' rather than just 'is it accurate?'.

A promise, and a catch

A model only learns what it is shown.

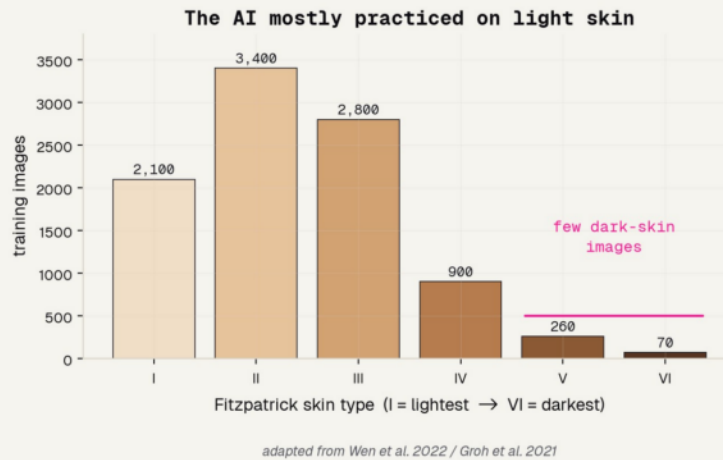
Skin-cancer AI is pitched as a way to reach the roughly three billion people with little access to a dermatologist. That promise is real, and so is the catch: the images these models learn from are badly skewed toward light skin.

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Slide 3 – A promise, and a catch

Skin-cancer AI could reach billions who never see a dermatologist, but a model only knows what its training images showed it, and those images skew heavily toward light skin. The promise and the risk are two sides of the same coin. Next we look at what the training data actually contains.

The AI mostly practiced on light skin

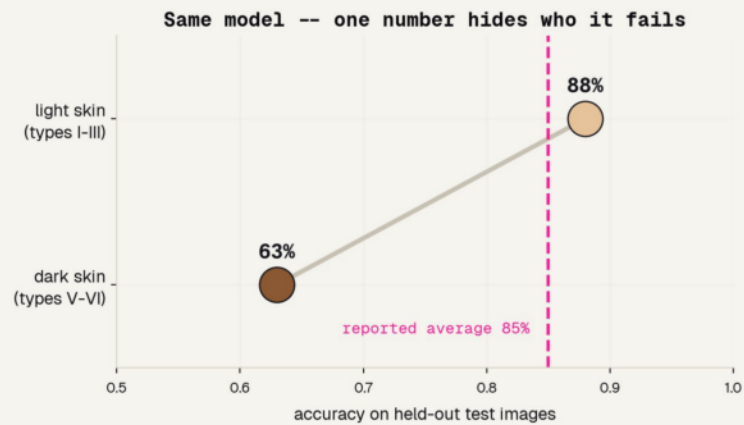


■ A review of twenty-one public skin datasets, over a hundred thousand images, found almost all came from Europe, North America, and Oceania, with almost none labeled as darker skin. On the Fitzpatrick scale the training data piles up on the left and nearly vanishes on the right.

Slide 4 – The AI mostly practiced on light skin

A systematic review of 21 public skin datasets (over 100,000 images) found almost all came from light-skinned patients in Europe, North America, and Oceania. On the Fitzpatrick I-VI scale the counts pile up on the lighter types and nearly vanish on the darker ones. A model that barely sees dark skin in training has little chance of doing well on it.

Same model, one number hides who it fails



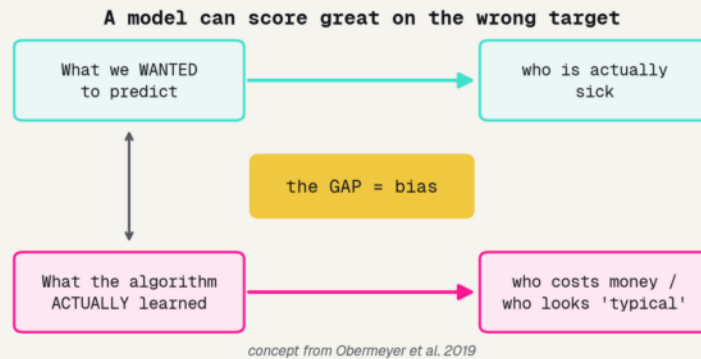
adapted from Daneshjou et al. 2022 / Groh et al. 2021

- Groh showed a model is most accurate on the skin types it saw most; Daneshjou showed top models drop sharply on dark skin. Same model, very different results depending on who you are, and the single average number a report shows sits in between and hides it.

Slide 5 – Same model, one number hides who it fails

Two studies make the point: Groh showed a model is most accurate on the skin types it saw most, and Daneshjou showed top models drop sharply on dark skin. The single average accuracy a report cites sits between the groups and hides the gap. This is why we distrust one headline number.

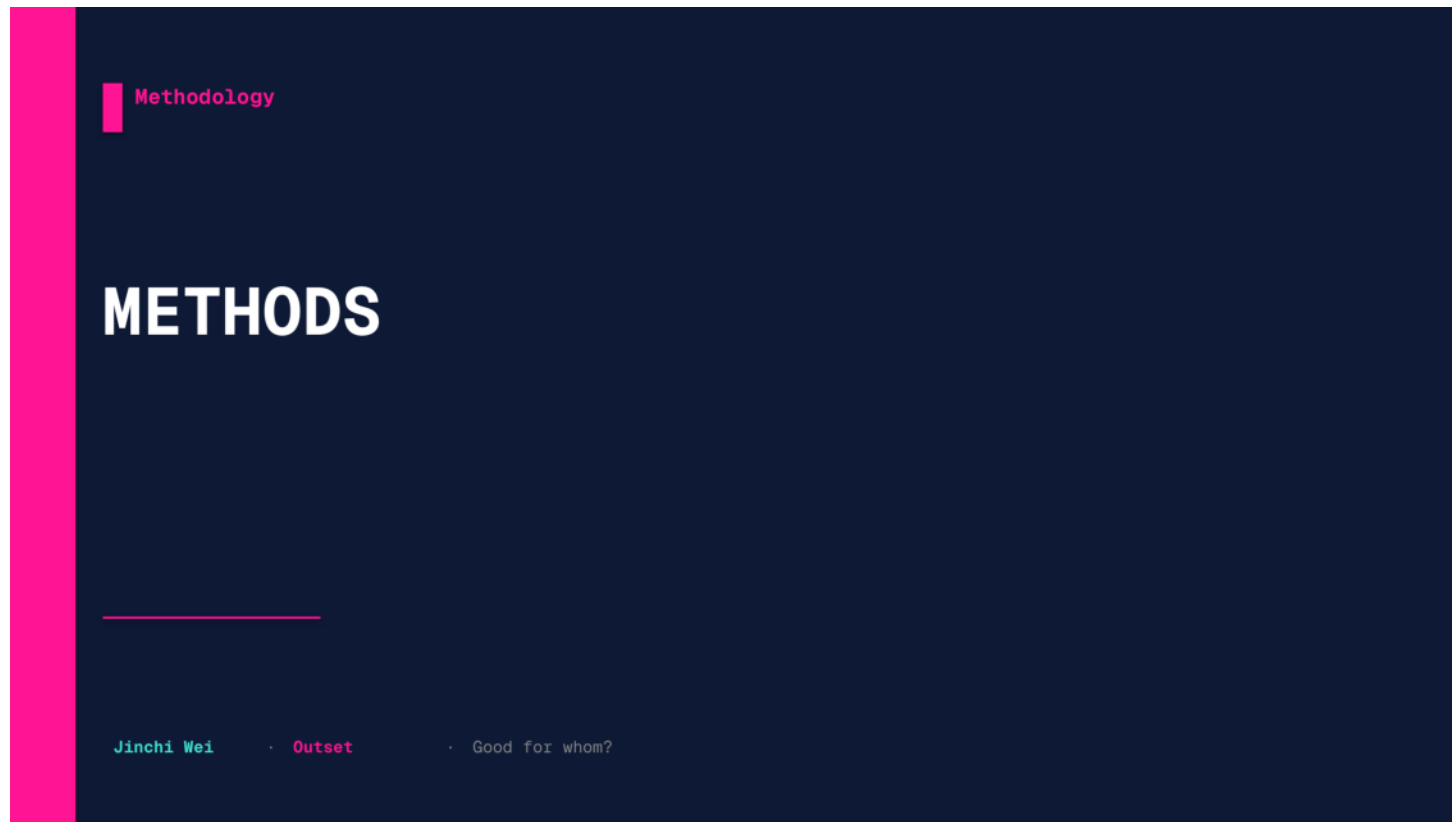
Accurate on average can still be unfair



- Obermeyer studied a health algorithm used on millions of patients that looked fair on paper but underserved Black patients, because it was trained to predict cost as a stand-in for illness. It scored well on the wrong target. That is the exact suspicion we bring to our own model.

Slide 6 – Accurate on average can still be unfair

Obermeyer's famous case: a health algorithm used on millions looked fair on average but underserved Black patients, because it predicted cost as a stand-in for illness -- the wrong target. A model can score well and still be unfair. That is exactly the suspicion we carry into our own model.



Slide 7 – Methods

Here we explain what we actually did. The key move is that we reuse the screening group's exact model and change only the question we ask of it -- from 'how accurate?' to 'accurate for whom?'.

The same model, a harder question

Change the question, not the model.

We reuse the screening group's exact DermaMNIST model -- a ResNet18 pretrained on ordinary photos, with a fresh head, sorting images into seven skin conditions. Same tool, harder question: not how good is it, but good for whom?

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Slide 8 – The same model, a harder question

We deliberately keep the screening group's model unchanged -- a ResNet18 with a fresh head sorting images into seven conditions -- so the only variable is the question. Same tool, harder question: not 'is it good?' but 'good for whom?'.

The equity lens: break the average apart

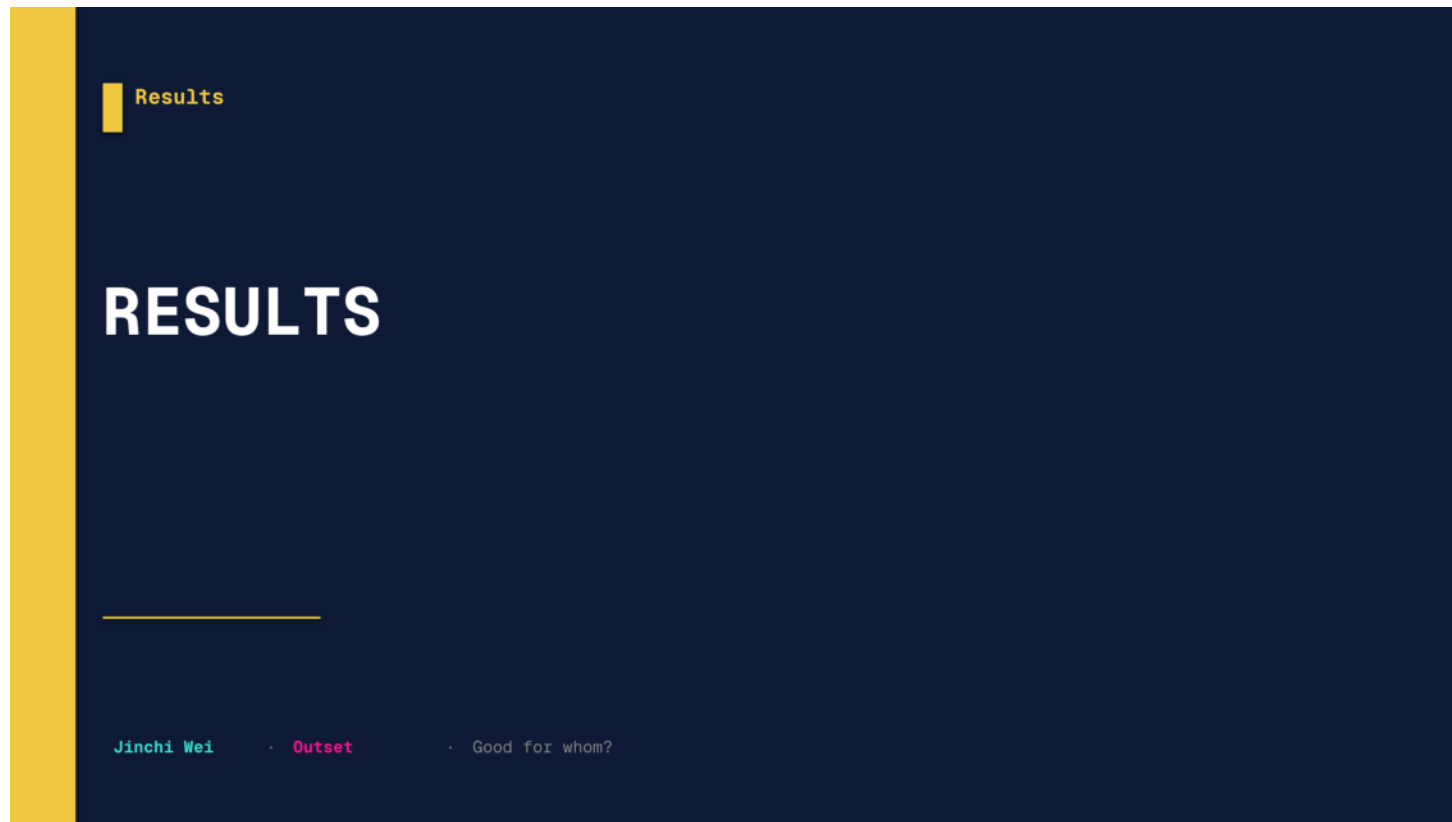
Good for whom?

A single accuracy number is a blend, and a blend can hide that one group is being carried by another. The screening group asks how good the model is; we refuse to trust the average and split the score apart by condition.

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Slide 9 – The equity lens: break the average apart

A single accuracy number is an average, and an average can let one strong group carry a failing one. The equity lens refuses the blend and splits the score apart. Here we split by condition, because that is the axis DermaMNIST actually labels.



Slide 10 – Results

Now we show what breaking the average apart reveals. Watch for the gap between the single headline number and the per-condition scores hiding beneath it.

The headline number looks fine

68%

overall accuracy

across all seven conditions, blended into one number

Not amazing, not embarrassing -- the kind of number you would quietly accept and move on. This is exactly the moment the equity lens says: stop, and break it apart.

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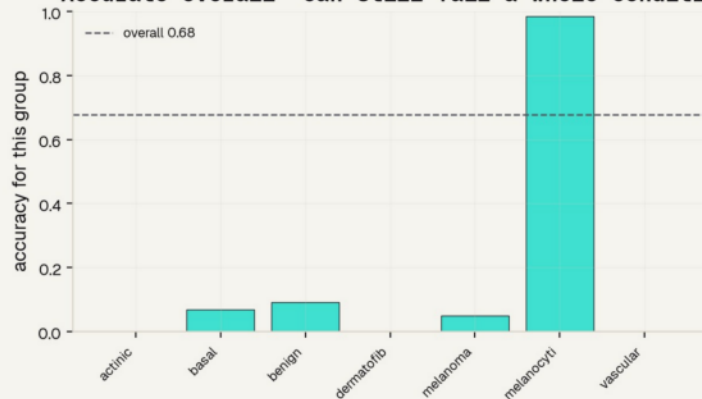
Slide 11 – The headline number looks fine

Overall the model scores about 68 percent -- unremarkable, the kind of number you would accept and move on.

That is precisely the moment to stop and disaggregate. The next slide breaks this number apart by condition.

Break it apart: a whole condition is failed

'Accurate overall' can still fail a whole condition



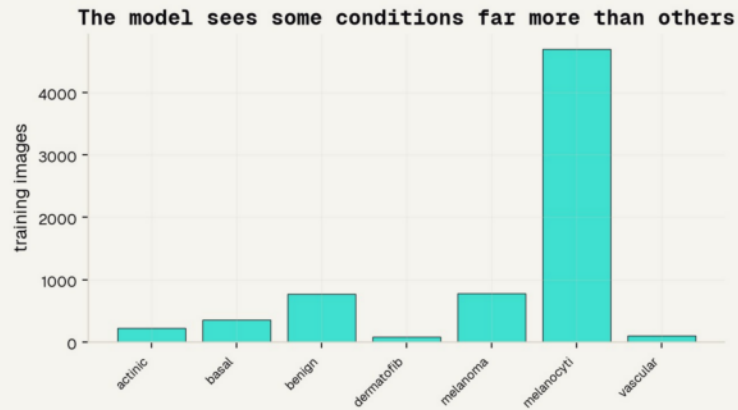
- Splitting the same score by condition, the average was hiding a near-total failure
- Ordinary moles, the giant class, are caught almost perfectly at ninety-nine percent and prop the average up. Several rarer conditions are caught almost never, some flat at zero.

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Slide 12 – Break it apart: a whole condition is failed

Splitting the 68 percent by condition exposes a near-total failure the average was hiding: common moles are caught about 99 percent of the time and prop the number up, while several rarer conditions are caught almost never, some at zero. The headline was carried almost entirely by one dominant class.

Why: the model sees some conditions far more



- The gap is not mysterious
- One class has thousands of training images and the rare conditions have only dozens. Whatever the model sees most, it gets best at. This is the same skew the literature describes for skin tone, along the axis DermaMNIST actually labels.

Slide 13 – Why: the model sees some conditions far more

The cause is simple: one class has thousands of training images while the rare conditions have only dozens, and a model gets best at whatever it sees most. This is the same imbalance the literature describes for skin tone -- shown here along the condition axis DermaMNIST records.

The blind spot we cannot measure

Who is in the training data?

DermaMNIST records the lesion type -- but not the patient's skin tone, age, sex, or country. Dermatology datasets skew heavily toward lighter skin.

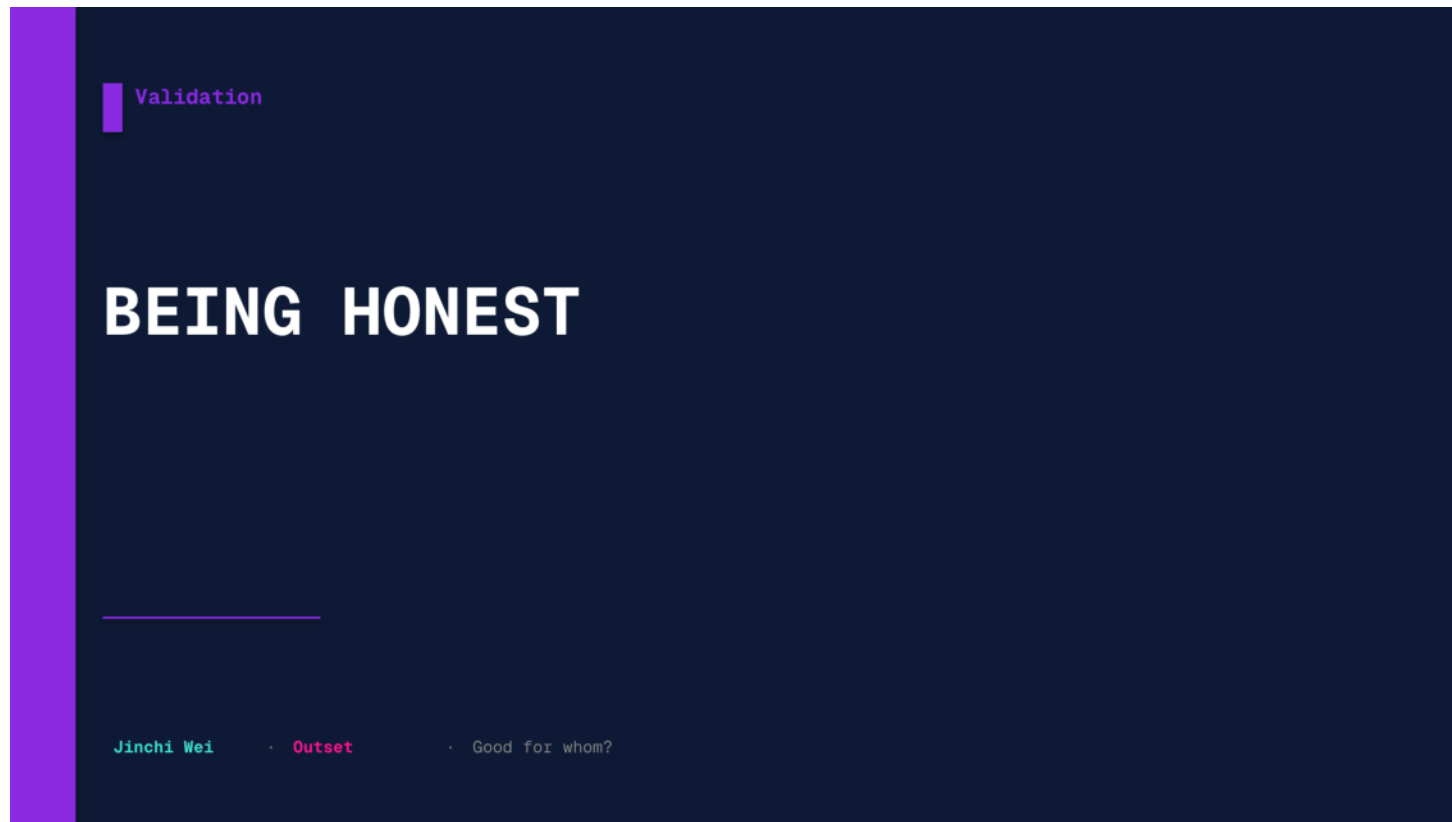
So we literally cannot check:

- ✗ Does it work as well on darker skin?
- ✗ Who was it never trained to see?
- ✗ Who gets helped -- and who gets missed?

- The gap the literature warns about most is along skin tone, and DermaMNIST ships no skin-tone labels. It records the lesion type and nothing about the patient. So we cannot check the real skin-tone gap here. We can demonstrate the risk of a blind spot, not measure it, and saying so is part of doing this honestly.

Slide 14 – The blind spot we cannot measure

The gap the literature worries about most is skin tone, but DermaMNIST ships no skin-tone labels -- only lesion type. So we can demonstrate the risk of a blind spot, not measure the real skin-tone gap. Saying that plainly is part of doing this honestly.



Slide 15 – Being honest

This closing section is about honesty: naming who the tool leaves out, and admitting what our dataset cannot tell us. Model the habit of stating limits out loud.

Who gets left out

Worst where care is scarcest.

The people the pitch was meant to help most are the ones most likely missing from the data -- disproportionately in regions and skin types barely present in training sets. A tool that is worst where care is scarcest can widen the gap it was sold to close.

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Slide 16 – Who gets left out

The cruel irony: the people the pitch was meant to help -- those with little access, often in underrepresented regions and skin types -- are the ones most likely missing from the training data. A tool that works worst where care is scarcest can widen the very gap it was sold to close.

References

1 Fitzpatrick 1988

The six-point skin-type scale (I-VI) dermatology AI now uses to describe who is in a dataset.

2 Groh et al. 2021

Fitzpatrick 17k: a model is most accurate on the skin types it saw most during training.

3 Daneshjou et al. 2022

Diverse Dermatology Images: state-of-the-art models drop sharply on dark skin tones.

4 Obermeyer et al. 2019

A health algorithm underserved Black patients by predicting cost instead of illness.

5 Wen et al. 2022

Review of 21 public skin datasets (106,950 images): almost all are light-skinned.

6 Adamson & Smith 2018

AI trained on fair-skinned archives can widen, not close, dermatology care gaps.

7 Buster et al. 2012

Skin-disease outcomes are worse for underserved, low-access, minority patients.

Slide 17 – References

These seven papers anchor the equity lens, from the Fitzpatrick scale that names the skin-type groups, to the studies that measured the accuracy gaps, to Obermeyer's proxy-bias case. Point students here for the primary sources.

Thank You

魏晉祺

Jinchi Wei

Outset

jinchikwei@gmail.com

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Slide 18 -